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DEPARTMENT OF COMPUTER ENGINEERING

EMOTION RECOGNITION FROM HUMAN VOICE SUPPORTED BY TEXT

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SENIOR PROJECT

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Güneşsu AÇIK
Osman Yiğit SÖKEL

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LIST OF ABBREVIATIONS

BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
CPU	Central Process Unit
CREMA-D	Crowd-sourced Emotional Multimodal Actors Dataset
CSV	Comma Separated Values
ESD	Emotional Speech Database
GeMAPS	Geneva Minimalistic Acoustic Parameter Set
GB	Gigabyte
GHz	GigaHertz
GPU	Graphics Processing Unit
HDD	Hard Disk Drive
IEMOCAP	Interactive Emotional Dyadic Motion Capture
LSTM	Long Short Term Memory
MELD	Multimodal EmotionLines Dataset
MFCC	Mel-Frequency Cepstral Co-efficients
ms	Millisecond
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
openSMILE	open-source Speech and Music Interpretation by Large-space Extraction
RAM	Random Access Memory
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
ReLU	Rectified Linear Unit
RoBERTa	Robustly Optimized BERT Pre-training Approach

RNN	Recurrent Neural Network
s	Second
SAVEE	Surrey Audio-Visual Expressed Emotion
SGD	Stochastic Gradient Descent
SSD	Solid State Drive
SVC	Support Vector Classification
SVM	Support Vector Machine
TESS	Toronto Emotional Speech Set
TB	Terabyte
TF-IDF	Term Frequency–Inverse Document Frequency
VADER	Valence Aware Dictionary for Sentiment Reasoning

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ABSTRACT

EMOTION RECOGNITION FROM HUMAN VOICE SUPPORTED BY TEXT

Güneşsu AÇIK
Osman Yiğit SÖKEL

Department of Computer Engineering
Senior Project

Advisor: Prof. Banu DİRİ

This study involves the design of a system that aims to recognize emotion from voice supported with text. Voice and text, both essential elements of communication, play a critical role in expressing people's emotional states. In light of this, the developed system takes the audio file and the text of this audio as the input, and extracts the emotion of the given data.

Within the scope of the project, detailed feature extractions were carried out to use voice and text features effectively. In order to process audio data, audio features were extracted with eGeMAPS of openSMILE library. Vectorization step was applied for text data. During the model development process, neural network-based approaches were preferred. The performance of the system has been increased by using CNN and deep learning algorithms. In this way, a more comprehensive emotion analysis was provided. In this process, the model was trained with a custom dataset created by merging multiple multimodal datasets. As a result, the system can recognize 5 different emotions: angry, sad, happy, surprised and neutral.

This study includes the stages of literature review, dataset processing, model determination, and training. This study contributes to a better understanding of emotion in communication technologies.

Keywords: emotion recognition, deep learning, neural networks, audio, text

ÖZET

YAZILI METİN İLE DESTEKLENMİŞ İNSAN SESİ ÜZERİNDEN DUYGU ÇIKARIMI

Güneşsu AÇIK
Osman Yiğit SÖKEL

Bilgisayar Mühendisliği Bölümü
Bitirme Projesi

Danışman: Prof. Dr. Banu DİRİ

Bu çalışma, ses ve metin kullanarak duygu analizi yapmayı amaçlayan bir sistemin oluşturulmasını içermektedir. Temel iletişim unsurlarından olan ses ve metin, insanların duygusal durumlarını belirtmede kritik bir rol oynamaktadır. Bunun doğrultusunda, geliştirilen sistem, girdi olarak alınan ses dosyası ve bu sese ait metinden duygu çıkarımı gerçekleştirmektedir.

Proje kapsamında, ses ve metin özelliklerinin etkili bir şekilde kullanılması için detaylı özellik çıkarımı gerçekleştirilmiştir. Ses verilerinin işlenmesi için openSMILE kütüphanesinden eGeMAPS seti kullanılarak özellikler çıkarılmıştır. Metin verileri için ise vektörize etme adımı uygulanmıştır. Model geliştirme sürecinde, sinir ağı temelli yaklaşımlar tercih edilmiştir. Özellikle CNN ve derin öğrenme algoritmaları kullanarak sistemin performansı arrtırılmıştır. Bu şekilde, daha kapsamlı bir duygu analizi sağlanmıştır. Bu süreçte model, birden fazla multimodal veri setinin birleşmesi sonucu oluşturulan veri seti ile eğitilmiştir. Sonuç olarak, sistem 5 duygu sınıfını tanıyabilmektedir: Sinirli, üzgün, mutlu, şaşkın ve normal.

Çalışma; literatür taraması, veri seti belirlenmesi, model belirlenmesi ve eğitimi aşamalarını içermektedir. Bu çalışma, iletişim teknolojilerinde duygunun daha iyi anlaşılmasına katkı sağlamaktadır.

Anahtar Kelimeler: duygu analizi, derin öğrenme, sinir ağları, ses, metin

1
Introduction

This chapter covers the need for the project, the project scope and the literature review.

1.1 Project Content

Over the last few years, rapid advancements in artificial intelligence and machine learning have led to important innovations in communications technology. One of these innovations is the improvement of the research on human voice and written texts. Communication fundamentally happens through interaction of voice and text. Therefore, these studies are carried out with the aim to gain a deeper understanding and analysis of these two base elements of communication.

This project aims to accurately define and understand human emotions while covering the fundamentals of communications technology. Combining voice and text data provides an environment to study and understand the emotional expressions in a more comprehensive way. Thus, communications technology can become more effective and human-oriented.

Internet significantly assists users for searching and accessing information. It is the primary tool used to get information about various products from various types of websites, such as e-commerce, hotel reservation, movie review, and other websites [1].

The tone of the voice delivers essential non-linguistic information and message, in addition the message delivered through text. This means same texts, sentences, words can have complete different meanings and show different emotions [2]. For instance, a sentence can have a positive meaning when expressed with a joyful tone, whereas the same sentence can have a negative meaning when expressed with a sorrowful tone. Therefore, vocalization of a text is critical to deliver the desired emotion and point. These emotions within the voice can provide more in-depth approach for emotion analysis.

In order to extract emotion from human voice and written text, neural network based approaches were used in this project using deep learning algorithms. Flexible structure of neural networks creates an appropriate environment to identify and understand emotional expressions.

1.2 Literature Review

Nowadays, thanks to the swift evolution of communications technology, a significant amount of research has been conducted on emotion recognition and analysis from human voice and written text. These studies aim to precisely understand and interpret emotional expressions, which is one of the main components of communication.

Within the scope of emotion analysis in voice, many studies have aimed to identify emotions using sound characteristics. SG Koolagudi and KS Rao (2012) have analyzed the existing studies and presented an extensive literature review, which compiled these studies to highlight the progress in this field [2]. Another study has focused on emotion recognition from music by associating emotions with music [3].

Research on sentiment analysis of written text can be associated closely with opinion mining. Sentiment analysis and opinion mining are studies that reveal emotions, opinions, and thoughts hidden in texts, which are expressed by people regarding different topics such as products services, organizations, events, etc [4]. These studies aim to detect the emotional content within texts.

In addition to all these research, studies on emotion recognition are carried out by using both voice and text data together. For instance, a study led by S.Yoon, S.Byun and K.Jung (2018) demonstrated sentiment analysis through the integration of audio and text data using RNN [5]. This research has surpassed the success of the previous studies.

In conclusion, as the technology evolves, there has been significant research in the field of sentiment analysis and emotion recognition. These studies have successfully achieved to obtain more effective and accurate results, contributing to precise analysis of emotional expressions.

2
Preliminary Examination

The emotion recognition system is designed for users to easily upload their data and learn about the emotions included within it, all through a user-friendly interface. Thus, this system can benefit individuals across a range of areas, including customer service, healthcare, education, call centers, and more.

The roadmap of the project consists of the following components:

1. Research for the compatible dataset, which includes both audio and text
2. Research and selection of the audio processing model suitable for the project
3. Research and selection of the text processing model suitable for the project
4. Integration of the processing models
5. Training the integrated model
6. Designing the interface of the system
7. Integration of all components
8. System improvements

3
Feasibility

In this chapter, the studies and requirements regarding the technical, time, economic, and legal feasibility of the project are explained.

3.1 Technical Feasibility

In this section, the software and hardware requirements of the project are evaluated.

3.1.1 Software Feasibility

The software language comparison to be used for the system software is shown in Table 3.1.

Table 3.1 Comparison of Software Languages

	Python	Java	C++
Ease of Use	Easy	Easy	Easy
Speed	Average	Average	Fast
Programmer Knowledge	Good	Good	Little
Machine Learning Libraries	Very Good	Average	Good
Interface Implementation	Good	Good	Little

- Machine Learning Libraries
  - Python; It has numerous libraries for machine learning. It can complete even complex operations more effectively with the built-in functions.
  - Java; Although it has various libraries for machine learning, it is not an effective language in this project. Because these libraries run slowly and take up a lot of space.
  - C++; Although it has machine learning libraries, its implementation is more complex than other languages.

- Interface Implementation
  - Python; Due to its libraries, fast and effective results are obtained.
  - Java; Although it seems like a good option with its built-in classes and libraries, it runs slower.
  - C++; Although it has libraries that support interface implementation, it is much more complex than other options. This complexity makes it difficult to get an effective result.

3.1.2 Hardware Feasibility

Project development processes were carried out through Google Colab with the two computers given in Table 3.2.

Table 3.2 Hardware Qualities

	Computer 1	Computer 2
RAM	32 GB	8 GB
Storage	256 GB SSD - 1 TB HDD	512 GB SSD
Processor	i7 10th Generation	Apple M1
Operating System	Windows 11	macOS Ventura

- Google Colab: Intel(R) Xeon(R) CPU @ 2.30GHz Nvidia Tesla V100 GPU 51 GB RAM 167 GB storage space

3.2 Time Feasibility

The Gantt Chart used for project planning is shown in Figure 3.1.

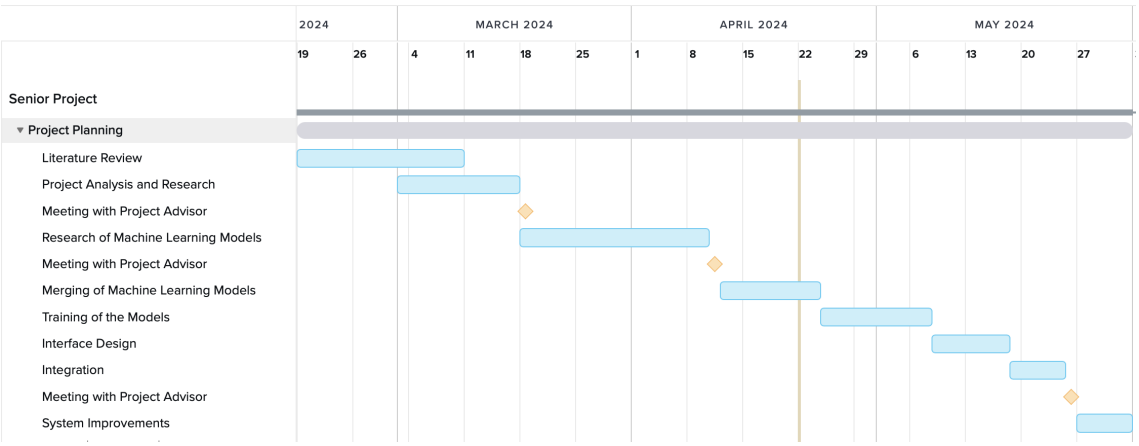

Figure 3.1 Gantt Chart

3.3 Economic Feasibility

The costs required for the project are given in Table 3.3.

Table 3.3 Cost of Software

	Feature	Cost
Google Colab	51 GB RAM 167 GB Storage Space NVIDIA T4 GPU	330 TL

All software technologies, libraries and applications used within the scope of this project are free. This allows the project to be developed in a more economically sustainable way.

3.4 Legal Feasibility

- **Libraries Used**: All libraries used in our project are open source and distributed under free software licenses. These licenses allow their users to freely use, distribute and modify the software. Open source libraries increase the sustainability of our project and provide a secure legal basis, as they are supported by a large developer community. Our project appropriately specifies the licenses of the libraries used and uses these libraries ethically.

- **Data Privacy and Protection of User Information**: User information is protected by modern security standards, including secure communication protocols, and is used only for specific purposes. Additionally, the open source libraries used in our project are constantly monitored and updated, representing a proactive approach to security vulnerabilities. In this way, our project provides effective protection in terms of user privacy and data security.

- **Illegal Content and Restrictions**: No illegal or restricted content is included in the data set.

4
System Analysis

In this chapter, system analysis and use-case scenario are explained.

4.1 Components of the System

- Hardware Resources
  - Computer
- Software Resources
  - Google Colab
- Human Resources
  - Güneşsu AÇIK | System Analyst and Programmer
  - Osman Yiğit SÖKEL | System Analyst and Programmer
  - User

4.2 Use-Case Scenario

Primary Actor: User
Stakeholders: System analysts

Basic Flow:

1. The user logs into the system.
2. The user gives the audio and text.
3. The emotion of the audio is shown to the user.

The use case diagram is included in Figure 4.1.

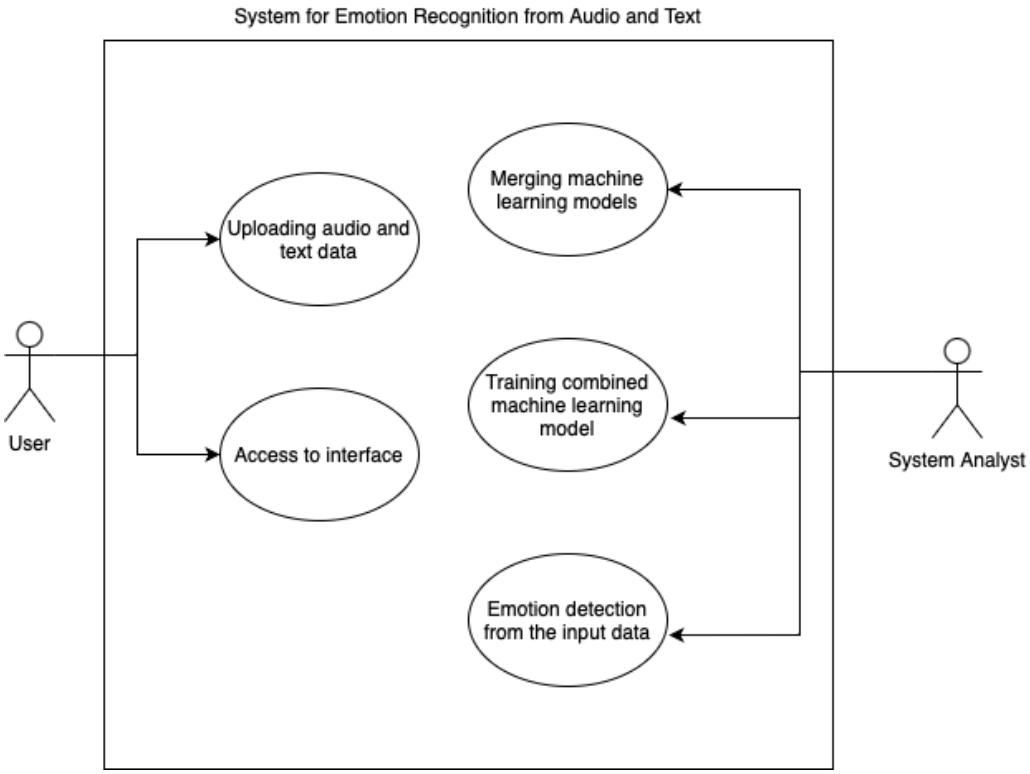

Figure 4.1 Use-Case Diagram

4.3 Data Flow Diagram

The use case diagram is included in Figure 4.2.

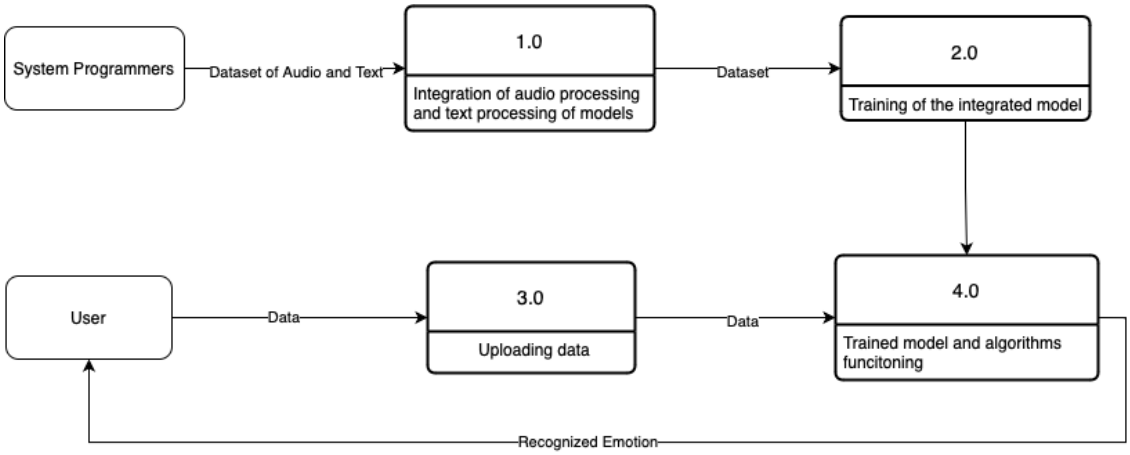

Figure 4.2 Data Flow Diagram

5
System Design

This chapter includes software, input-output, database design and the research, comparisons and analysis carried out throughout these stages.

5.1 Software Design

5.1.1 Dataset Processing

Multimodel datasets include both voice and text data, which allows us to perform sentiment analysis and emotion extraction more comprehensively. This approach creates a better and more in-depth analysis by the integration of data from diverse sources.

5.1.2 Determination of Audio Model

This section covers the use of a voice model for emotion recognition. Our main goal is to develop a deep learning model that extracts emotions from human voice. In pursuit of this goal several voice models were evaluated.

5.1.2.1 Dataset Processing for Audio Model

CREMA-D [6], RAVDESS [7], SAVEE [8], and TESS [9] datasets were used to create a more inclusive, integrated dataset. This integration process aims to combine the features of the data sets in a homogeneous way and make the dataset suitable for analysis.

Same labels were used in all datasets, which simplified the process of merging and comparison of datasets. Emotion distribution of the merged dataset is given in Figure 5.1.

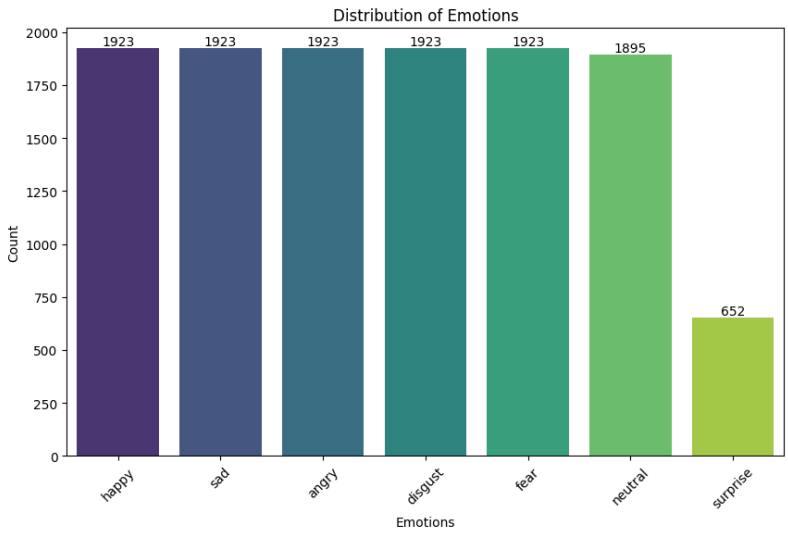

Figure 5.1 Distribution of Emotions/Speech

5.1.2.2 CNN Model

CNN is a class of artificial neural networks, commonly used in computer vision tasks owing to its high success rates. Due to its design it can easily and adaptively learn spatial hierarchies of features through back-propagation by using multiple building blocks. This operation includes convolution layers, pooling layers, and fully connected layers [10].

CNN operates on spectrograms in the context of emotion recognition in audio data, utilizing convolutional and pooling layers. These layers automatically extract features such as pitch and intensity, facilitating emotion classification. The parallel computation and weight sharing advantages of CNNs deliver fast and effective results on datasets. Consequently, the model analyzes the emotional content in audio data and identifies different emotional states. CNN plays a significant role in emotion recognition, leveraging deep learning techniques.

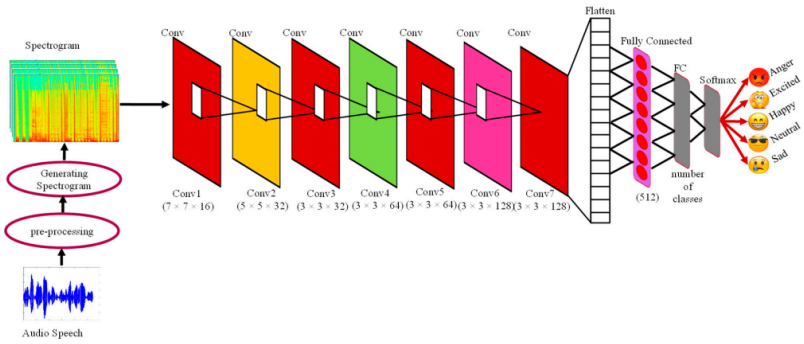

Figure 5.2 Architecture of CNN for Speech Emotion Recognition [11]

5.1.2.3 LSTM Model

Being a modified RNN method, LSTM works well on a large variety of problems, and is widely used now [12]. Unlike traditional RNN models, LTSM has a cell structure that can preserve long-term dependencies effectively. This cell has a memory unit called cell state, which consistently screens and updates information.

In addition, LSTM uses gate mechanism where these gates control the flow of information to the memory cell. This is done by generating a value between 0 and 1. This gate mechanism helps the model to learn and perform complicated tasks such as sentiment analysis.

Thanks to these qualities, LSTM has been successfully used in various fields, including voice data processing such as emotion recognition. Analysis of emotional expressions from voice data requires understanding of long-term dependencies and LSTM outshines with its ability to learn these type of complex relations. The use of LSTM in voice models allows us to obtain precise results. LSTM architecture is given in Figure 5.3.

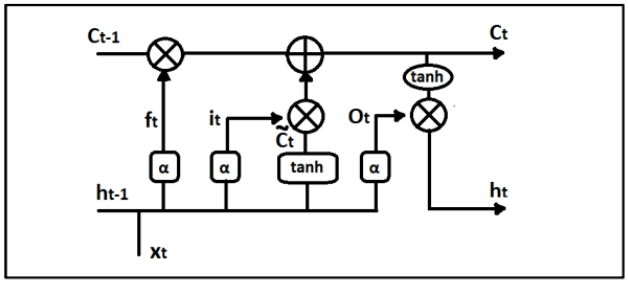

Figure 5.3 LSTM Structure
[13]

5.1.2.4 Comparison and Conclusion

CNN and LSTM based deep learning models were trained with a sample dataset to compare their performances. The purpose of this comparison is to decide the most suitable model that meets the needs for the project. The comparison of these models is given in Table 5.1. Accuracy/loss values of LSTM and CNN are given in Figures 5.4 and 5.5 in order.

Table 5.1 Comparison of Voice Model Results

	LSTM Model	CNN Model
Training Time	450ms/epoch	325ms/epoch
Accuracy	0.89	0.96
Loss	0.32	0.13

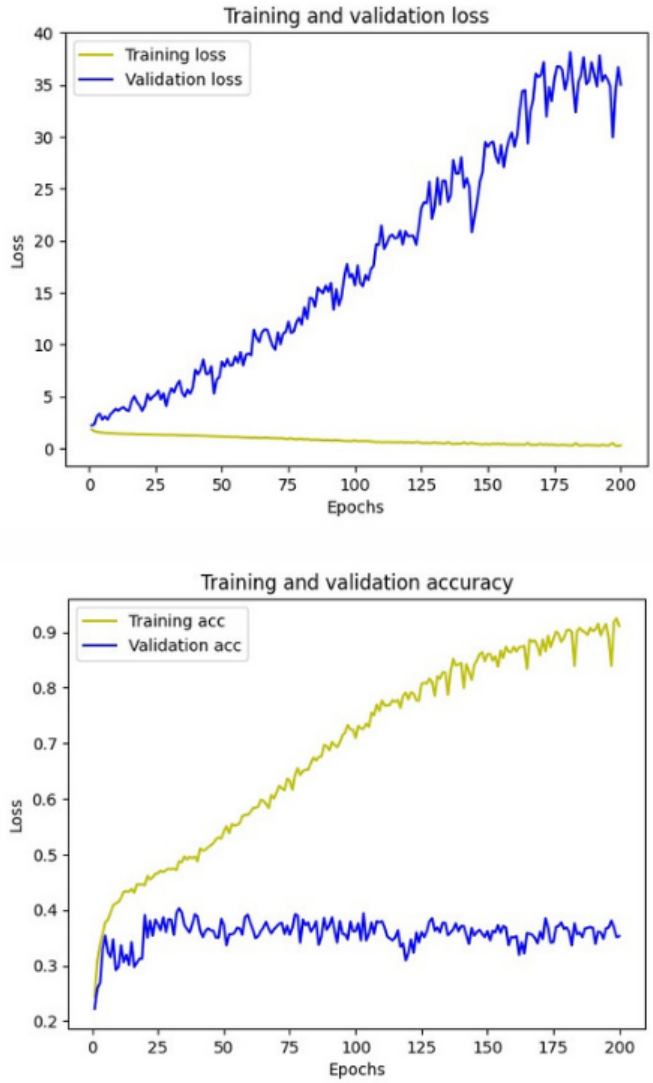

Figure 5.4 Accuracy/Loss of LSTM

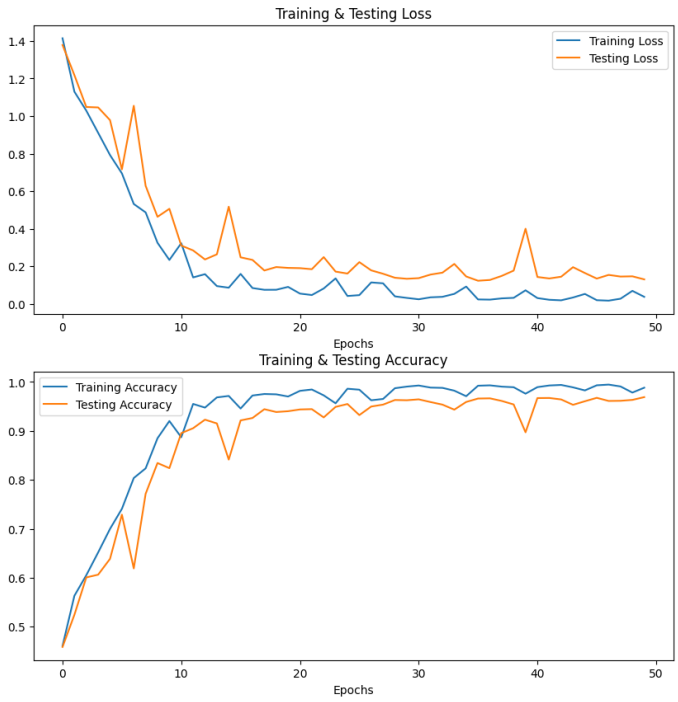

Figure 5.5 Accuracy/Loss of CNN

5.1.3 Determination of Text Model

In the project, achieving optimal accuracy in output was our top priority. Therefore, we thoroughly tested multiple modules to ensure the highest level of accuracy. Research findings and results from these modules are presented below.

5.1.3.1 Dataset Processing for Text Model

A test dataset was used to determine the success of each module. This dataset covers 7 emotions and the distribution of these emotions is given in Figure 5.6

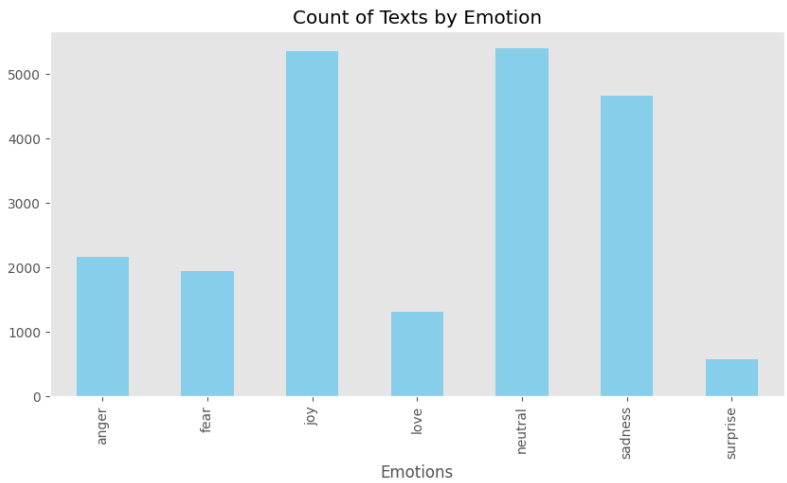

Figure 5.6 Distribution of Emotions/Text

5.1.3.2 Natural Language Toolkit

Natural Language Toolkit is a Python suite of libraries for natural language processing. This library, can roughly extract the emotion via obtaining the emotional score of the sentence and comparing it with the given label. NLTK includes graphical demonstrations and sample data, which makes it suitable for a research project [14]. Working principle of a simple information extraction system, where NLTK's default functions are used throughout these steps, is shown in Figure 5.7

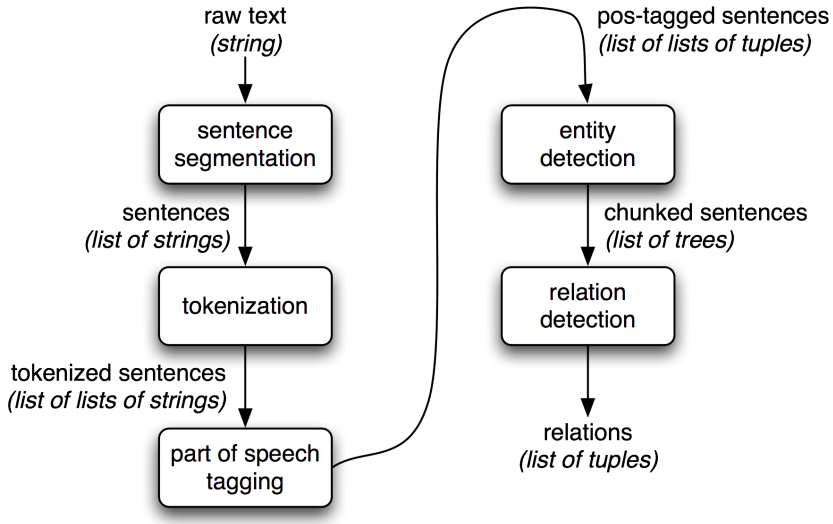

Figure 5.7 Working Principle of Information Extraction System [15]

5.1.3.3 VADER

VADER is an NLTK module, which requires no training data, but is constructed from a generalizable, valence-based standard sentiment lexicon [16]. It detects how positive or negative every word is in a sentence to determine the overall sentiment of a given text. Tokenization and chunking for this process is given in Figure 5.8.

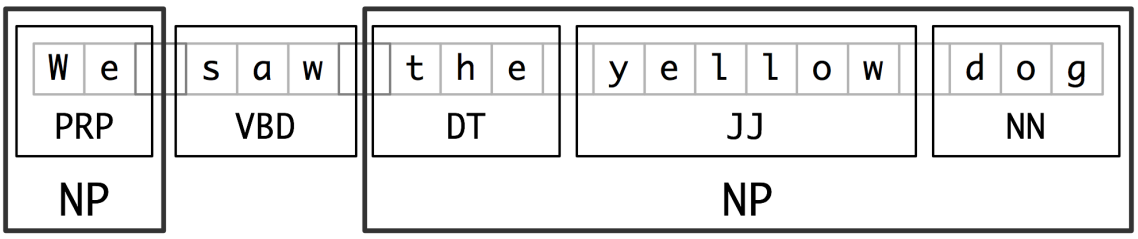

Figure 5.8 Tokenization and Chunking

5.1.3.4 RoBERTa

RoBERTa, with its larger training data and smarter training methods, is an extension of BERT. BERT, and therefore RoBERTa, stands out from other NLP models since they can learn contextual representation from both ends of the sentences [17]. Both of these models are transformer models. These models consist of tokenizer, transformer, and heads. For the tokenization part, RoBERTa uses a byte-level Byte-Pair Encoding with a 50k word vocabulary [18]. Architecture of RoBERTa is given in Figure 5.9

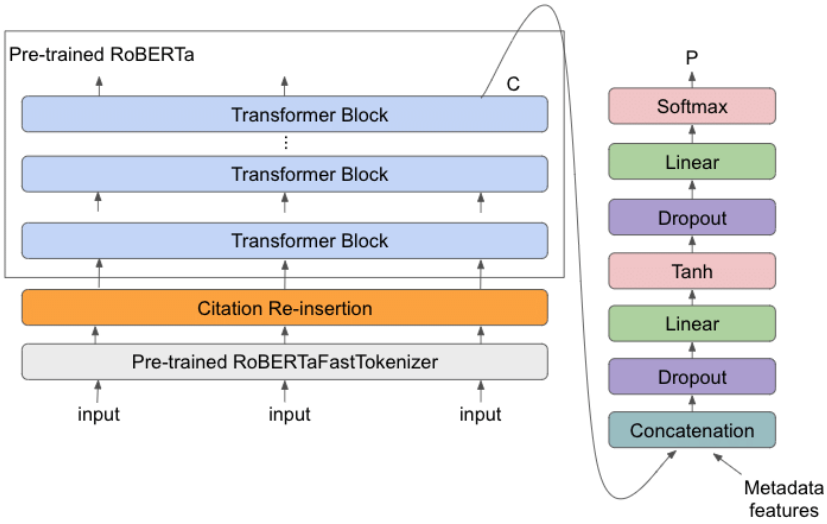

Figure 5.9 Architecture of RoBERTa Model
[19]

5.1.3.5 Support Vector Machine

SVM is one of the classical machine learning techniques that can still be used for current problems [20]. SVC is an implementation of SVM, specifically used for classification problems. The architecture of SVM is given in Figure 5.10

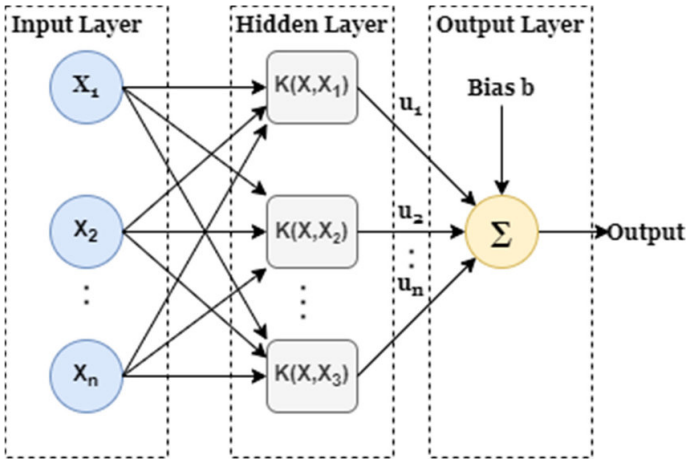

Figure 5.10 Architecture of SVM
[21]

5.1.3.6 Integrated Text Model

VADER and RoBERTa are both pre-trained modules and can produce a high level success on given dataset. The scatter plot to compare and analyze the correlation of VADER and RoBERTa is given in Figure 5.11.

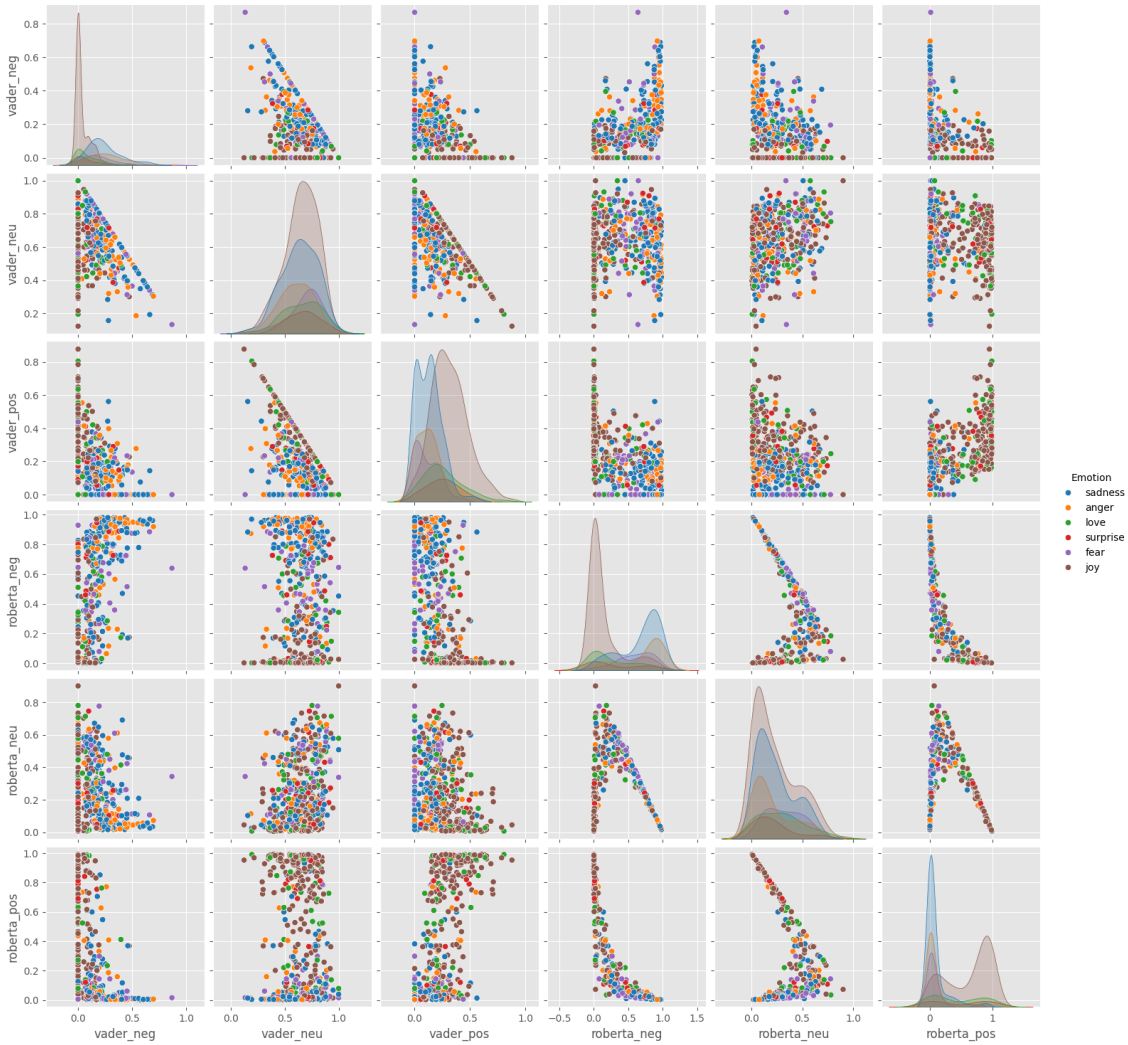

Figure 5.11 Scatter Plot of VADER and RoBERTa

In addition to VADER and RoBERTa's success, we have achieved a high level of success from the training of SVC. All this is taken into consideration, we tried training SVC supported with VADER and RoBERTa in order to achieve a higher success rate for text model. However, the accuracy decreased from 0.86 to 0.73. Therefore, we have decided to use SVC alone, as the text part of the integrated model.

5.1.3.7 Comparison and Conclusion

NLTK modules and SVC have been used for training with a test dataset. The results of these training are given in Table 5.2 and 5.3. VADER and RoBERTa are pre-trained

models, therefore there has been no training for those modules.

Table 5.2 NLTK Module Training Results

	Train Data	Test Data
Accuracy	0.9957	0.8725
Loss	0.0142	0.7180

Table 5.3 Classification Report of SVC

	Precision	Recall	F1-Score	Support
Anger	0.90	0.71	0.79	475
Fear	0.78	0.72	0.75	378
Joy	0.76	0.92	0.83	1051
Love	0.82	0.70	0.75	254
Neutral	0.96	0.98	0.97	1024
Sadness	0.90	0.85	0.87	981
Surprise	0.81	0.61	0.69	117
Accuracy	-	-	0.86	4280
Macro Average	0.85	0.78	0.81	4280
Weighted Average	0.86	0.86	0.85	4280

5.1.4 Determination of Feature Extraction Method

5.1.4.1 Librosa

Librosa is an audio analysis library designed with Python. It offers a wide range of toolkit for audio features and signals and it is generally used for audio processing. We used Librosa for feature extraction from audio files in this study. The extracted features are given below.

- **Spectral Centroid:** Center of mass of the sound spectrum
- **Spectral Bandwidth:** Width of the energy distribution of the sound spectral
- **Spectral Contrast:** Wave height of the sound spectrum
- **Spectral Flatness:** Intonation of the audio signal
- **ZCR:** Number of places where the signal crosses the zero axis
- **RMSE:** Energy level of the signal

- **Tonnetz:** Analysis of the harmonic properties of the sound
- **Tempogram:** Analysis of the tempo changes of the signal
- **MFCC:** Optimization of the sound for human ear
- **Chroma:** Representation of the audio signal based on the musical notes

In order to better use of these features,some statistical calculations were made such as, mean, standard deviation, skewness, median, minimum, and maximum.

A general feature matrix was created with these statistical calculations. In total, 239 features were extracted for each data. The accuracy and loss of the model, which is trained using these extracted features, are given in Figure 5.12. Classification report and confusion matrix are also given in Table 5.4 and Figure 5.13.

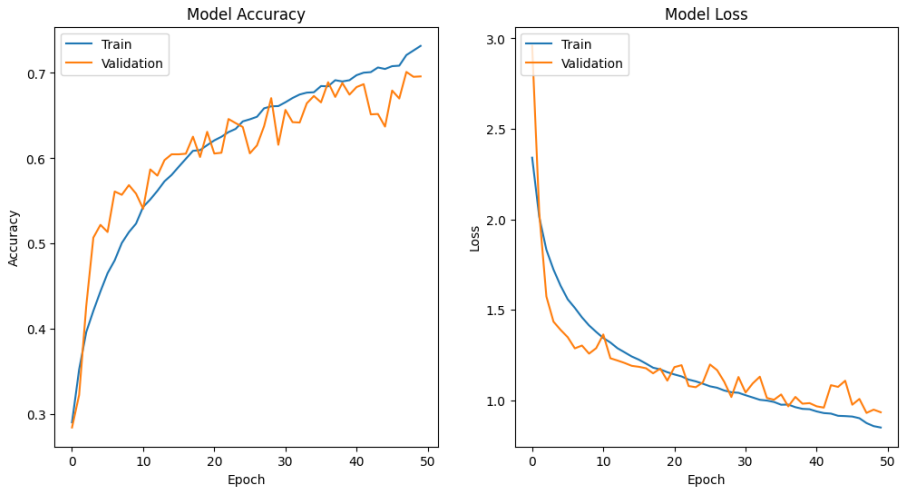

Figure 5.12 Accuracy/Loss of Librosa

Table 5.4 Classification Report of Librosa

	Precision	Recall	F1-Score	Support
Anger	0.65	0.75	0.69	1483
Happy	0.71	0.54	0.59	1571
Neutral	0.62	0.81	0.70	2454
Sad	0.79	0.64	0.71	1969
Surprise	0.71	0.59	0.65	1330
Accuracy	0.68	0.68	0.68	8807
Macro Average	0.70	0.67	0.67	8807
Weighted Average	0.69	0.68	0.68	8807

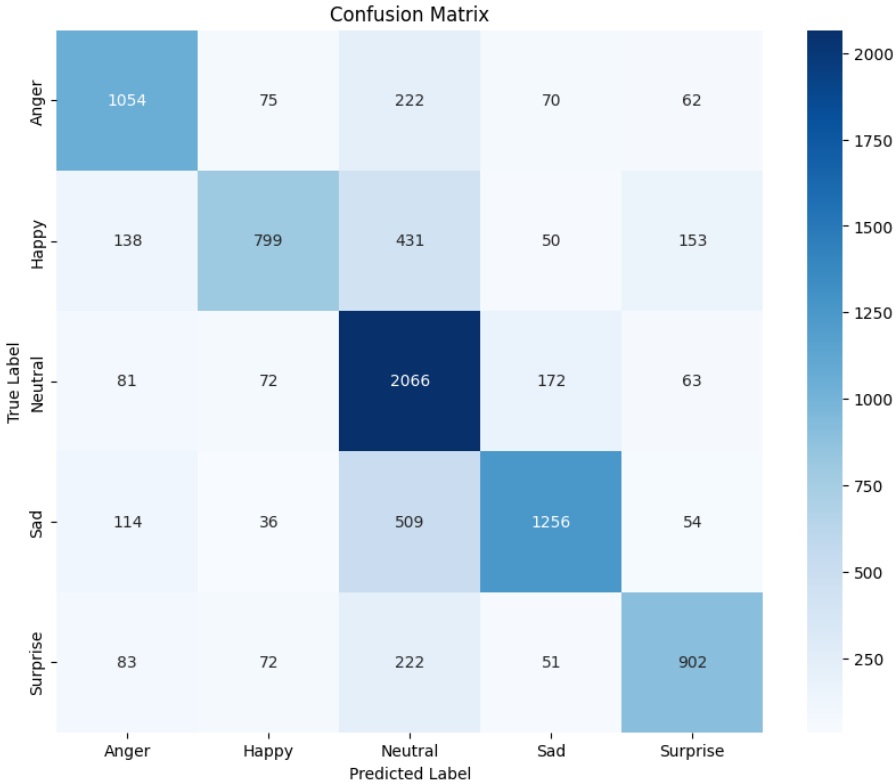

Figure 5.13 Confusion Matrix of Librosa

Although we have achieved high level of success during training step with features extracted with Librosa, success rate with test dataset was below expectations.

5.1.4.2 openSMILE

openSMILE, an audio feature extraction algorithm, was used for the project. openSMILE's supported audio low-level descriptors include waveform, signal energy loudness, FFT spectrum ACF, cepstrum Mel/Bark spectrum, Semitone spectrum, cepstral, pitch, voice quality LPC, auditory formants, spectral, tonal [22]. Architecture of openSMILE is given in Figure 5.14.

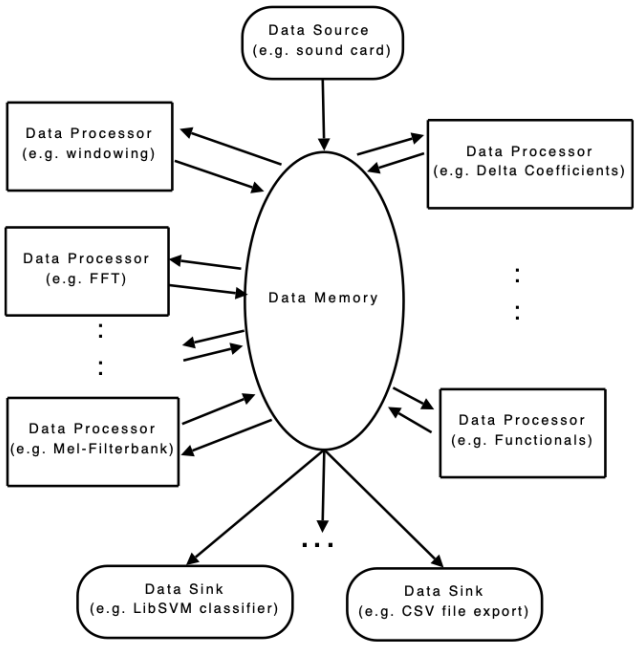

Figure 5.14 openSMILE Architecture
[22]

Features extracted with openSMILE achieved a high level of success with training and test dataset. Therefore, we have decided to move forward with openSMILE for feature extraction step.

5.1.5 Interface Design

This project aims to contribute to ongoing studies on multimodal emotion recognition systems and is not intended for user experience. The success of the project is evaluated based on accuracy scores. Therefore, an interface is not necessary for the project.

5.2 Input-Output Design

The system provides the chance to learn the dominant emotion in voice recordings by uploading both the voice and text. The machine learning module we've developed analyzes this data and displays the extracted emotion.

5.3 Database Design

A database design based on ready-made datasets, for the use of machine learning models, was used for this project. These datasets are stored in the Google Drive folder, providing accessibility and organization.

6
Application

In this section, the implementation and integration of the studies are explained.

6.1 Application of Voice Processing Model

6.1.1 Feature Extraction from Audio Files

In this step, noise, stretch, shift, and pitch functions were created in order to process audio files under various conditions. Next, the focus was on feature extraction from audio files. These features include:

1. **zcr:** Computes the zero-crossing rate. This provides information about the fluctuation and rate of change in the audio signal.
2. **rmse:** Computes the Root Mean Square Energy. This measures the overall energy level of the audio signal.
3. **mfcc:** Computes Mel-frequency cepstral co-efficients. It's an attribute used to understand and analyze the frequency content of an audio signal.

6.1.1.1 MFCC

MFCC is a feature extraction technique that has a system similar to human auditory frequency sensitivity. It splits audio data into its components and converts them into processed data.

Thanks to this ability, it is used in various areas such as, speech recognition, emotion recognition, music classification, and more. MFCC algorithm is given in Figure 6.1.

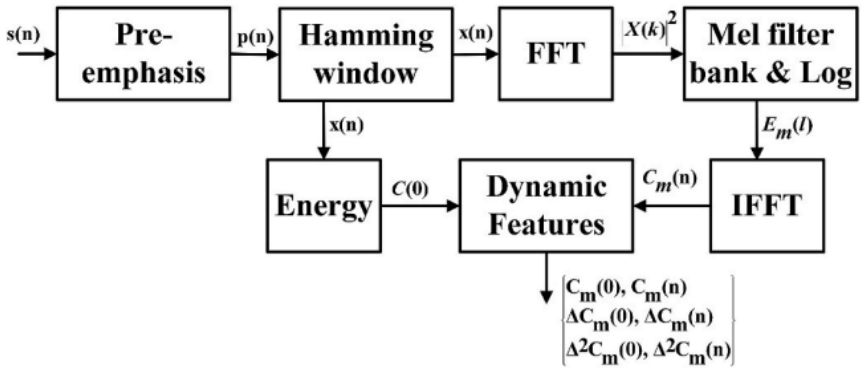

Figure 6.1 Mathematical model of MFCC algorithm
[23]

6.1.2 Application of CNN Model

Labels are converted into binary coding with 'OneHotEncoder', which helps with multi-label classification. Dataset is split into train and test parts.

The model uses CNN to process voice signal data. This divides the data into small pieces, converts these pieces into feature vectors, and uses artificial neural network for classification.

Confusion matrix and report table is obtained after the training of the model. Confusion matrix, given in Figure 6.2, evaluates the predictive performance of the model. Reports table, given in Table 6.1, displays precision, recall, F1-score values of the model.

Table 6.1 Classification Report of CNN

	Precision	Recall	F1-Score	Support
Angry	0.96	0.97	0.96	1525.0
Disgust	0.96	0.96	0.96	1580.0
Fear	0.96	0.95	0.96	1530.0
Happy	0.96	0.97	0.97	1475.0
Neutral	0.97	0.96	0.97	1564.0
Sad	0.96	0.97	0.96	1561.0
Surprise	0.99	0.97	0.98	495.0
Accuracy	0.96	0.96	0.96	9730.0
Macro Average	0.97	0.96	0.97	9730.0
Weighted Average	0.96	0.96	0.96	9730.0

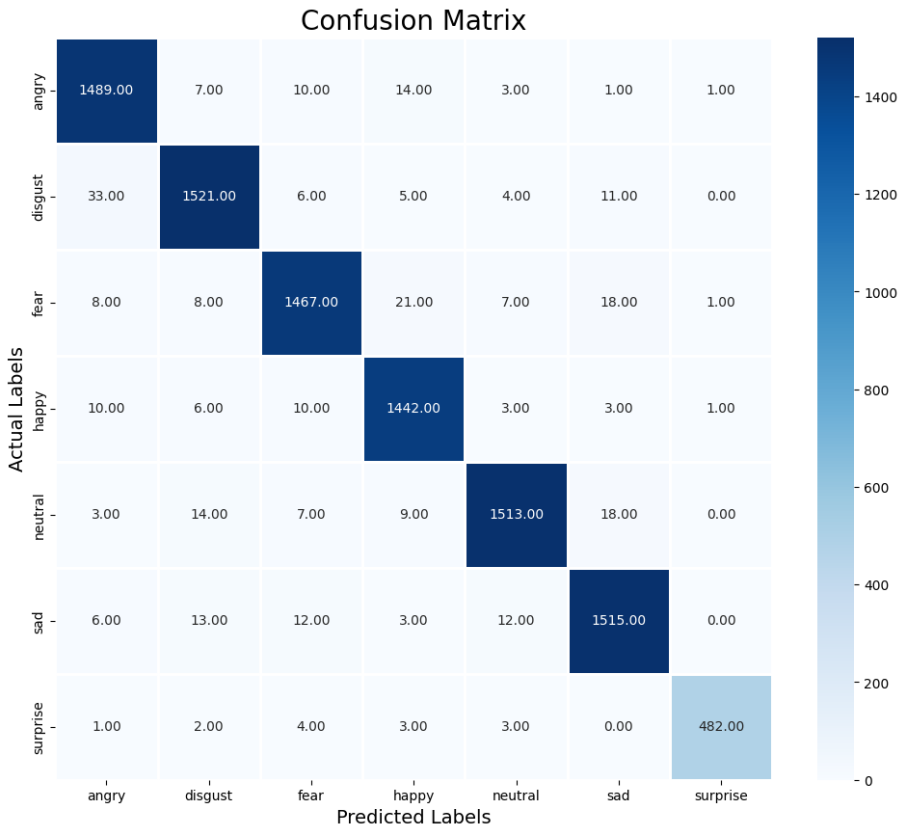

Figure 6.2 Confusion Matrix of CNN

6.2 Application of Text Processing Model

6.2.1 TF-IDF Vectorization and Label Encoding

TF-IDF is a calculation of relevance of a word in a given text. Fundamentally, the algorithm measures the originality of each word with the formula given below.

For term x within document y:

$$w_{x,y} = \mathrm{tf}_{x,y} \times \log\left(\frac{N}{\mathrm{df}_x}\right)$$

where,

$$\begin{aligned} \mathrm{tf}_{x,y} &= \text{frequency of } x \text{ in } y \\ \mathrm{df}_x &= \text{number of documents containing } x \\ N &= \text{total number of documents} \end{aligned}$$

This step transforms words into a meaningful representation of numbers, preparing them for machine learning models. Architecture of TF-IDF is given in Figure 6.3.

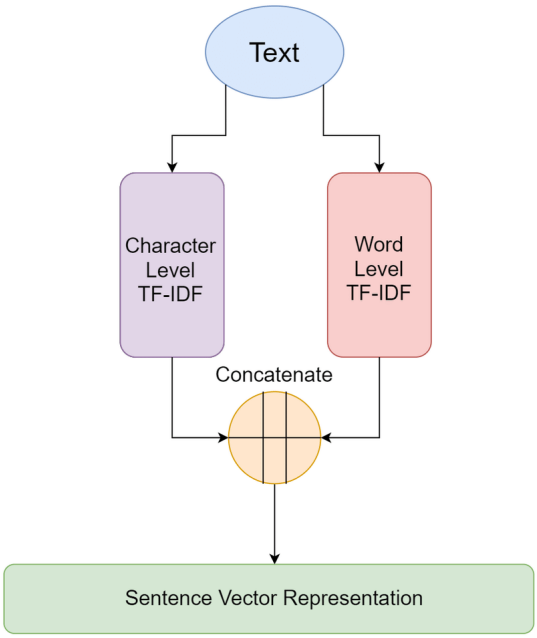

Figure 6.3 TF-IDF Architecture
[24]

For the next step, we used LabelEncoder from Scikit-learn library of Python to convert emotion labels into numerical format. This is an initial step for the machine learning model.

6.2.2 Application of SVM

We used incremental learning with SVM, in order to use the memory efficiently. First the pipeline was set up. During this step we used StandardScaler and SGDClassifier, both from Scikit-learn library. StandardScaler removes the mean and scales each feature to unit variance. This tep enhances the model performance. SGDClassifier is a linear classification algorithm which operates by iteratively updating the model's parameters using the SGD optimization technique. Working principle of incremental learning is given in Figure 6.4.

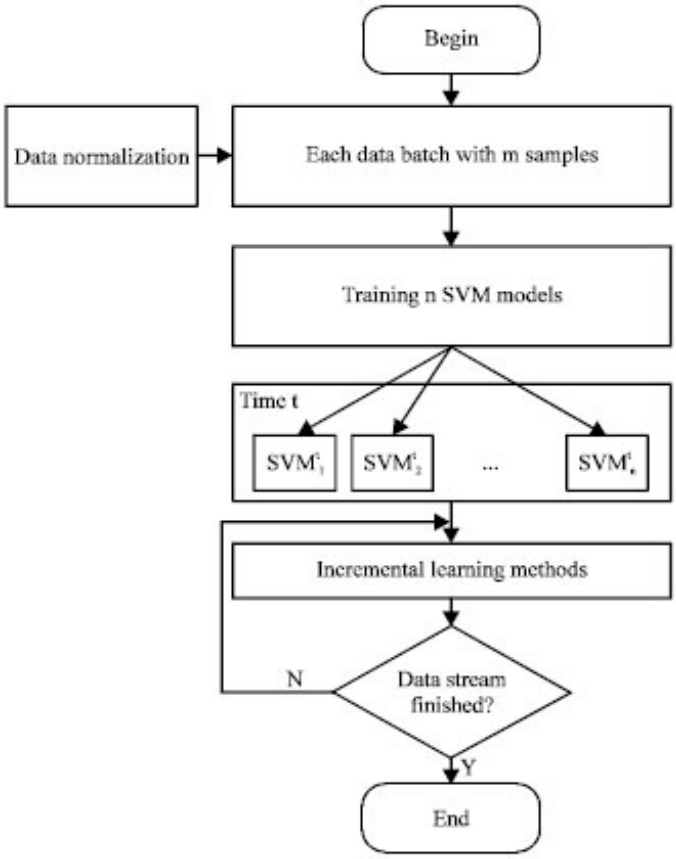

Figure 6.4 Incremental Learning
[21]

6.3 Application of Integrated Model

6.3.1 Dataset Pre-processing

The dataset used for emotion analysis on this project was a compilation of different multimodal datasets including IEMOCAP, CREMA-D, MELD, RAVDESS, ESD and JL-Corpus. The data in these datasets were merged on a single data matrix and saved to a CSV file. This new and merged dataset includes file paths, texts of the audios, and emotion labels.

Some operations were performed to ensure balance in the data set. Labels of emotions within the similar range are collected under a common label. Emotion labels that were very few in number were removed from the dataset. Additionally, files that has problems in their audio paths have been removed. After all these operations, the final distribution of the dataset is given in Figure 6.5.

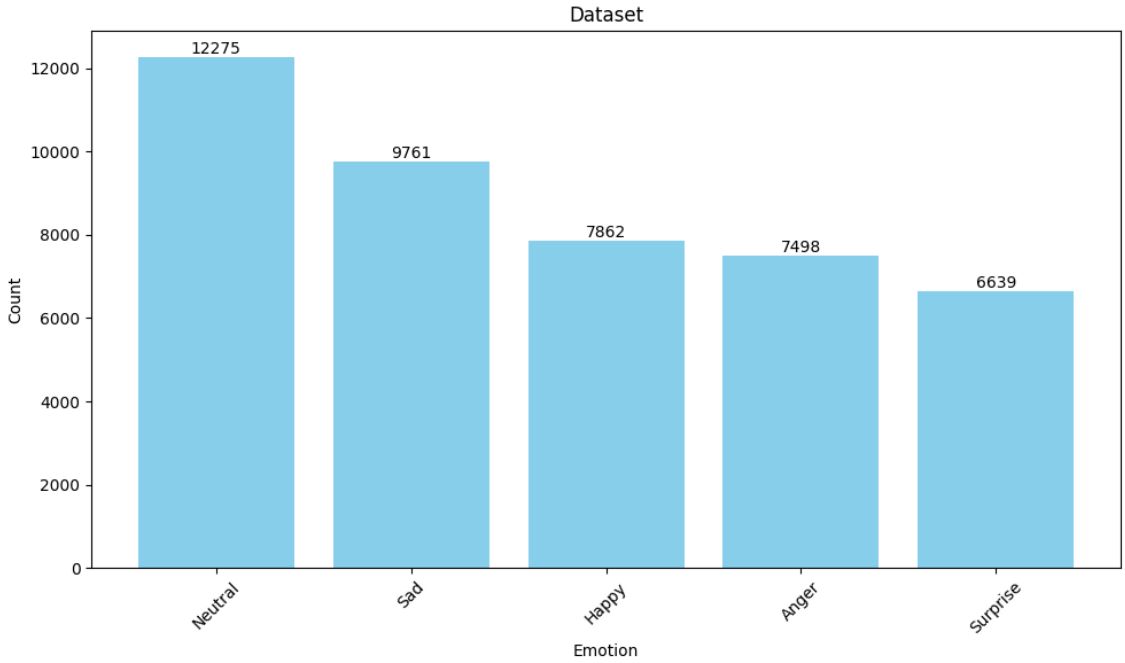

Figure 6.5 Distribution of Emotions

For the following step of the project, features were extracted from audio files using Librosa library. Text data were vectorized with TF-IDF. The 1000 most frequently used features were extracted from the text data. An SVM model was trained using the obtained TF-IDF vectors. In this way, probability estimates were obtained for text data. Features extracted from audio and text files are combined and scaled.

6.3.2 Feature Extraction

For the feature extraction part, we used extended GeMAPS version 2 from openSMILE with the following piece of code.

```
smile = opensmile.Smile(
    feature_set=opensmile.FeatureSet.eGeMAPSv02,
    feature_level=opensmile.FeatureLevel.Functionals,
)
```

GeMAPS set includes 18 low-level descriptors and 6 temporal features. Extended version includes MFCC 1–4, spectral flux differences, formant 2–3 bandwidth in addition to those descriptors and features [25].

6.3.3 Model

For this stage of the project, the features extracted from audio and text was combined together. Combined features were scaled within the range of [0,1] with MinMaxScaler. Thus, the performance of the model was increased by ensuring that data at different scales were in the same range.

In the early stages of the project, a CNN model was created to analyze emotion from audio data. Also, a layer with 512 filters was added for text data. Combined features were given as input to the integrated model. Architecture of the model consists of convolutional layer, pooling layer, batch normalization layer, and droupout layers. Convolutional layer extracts features from input data. Batch normalization layer is used to increase the stability of the model. Max pooling layer reduces the computational cost of the model by reducing spatial dimensions. Droupout layers prevents overfitting. Lastly, classification layer was created by adding softmax activation. The visuliation of the model is given in Figure 6.6.

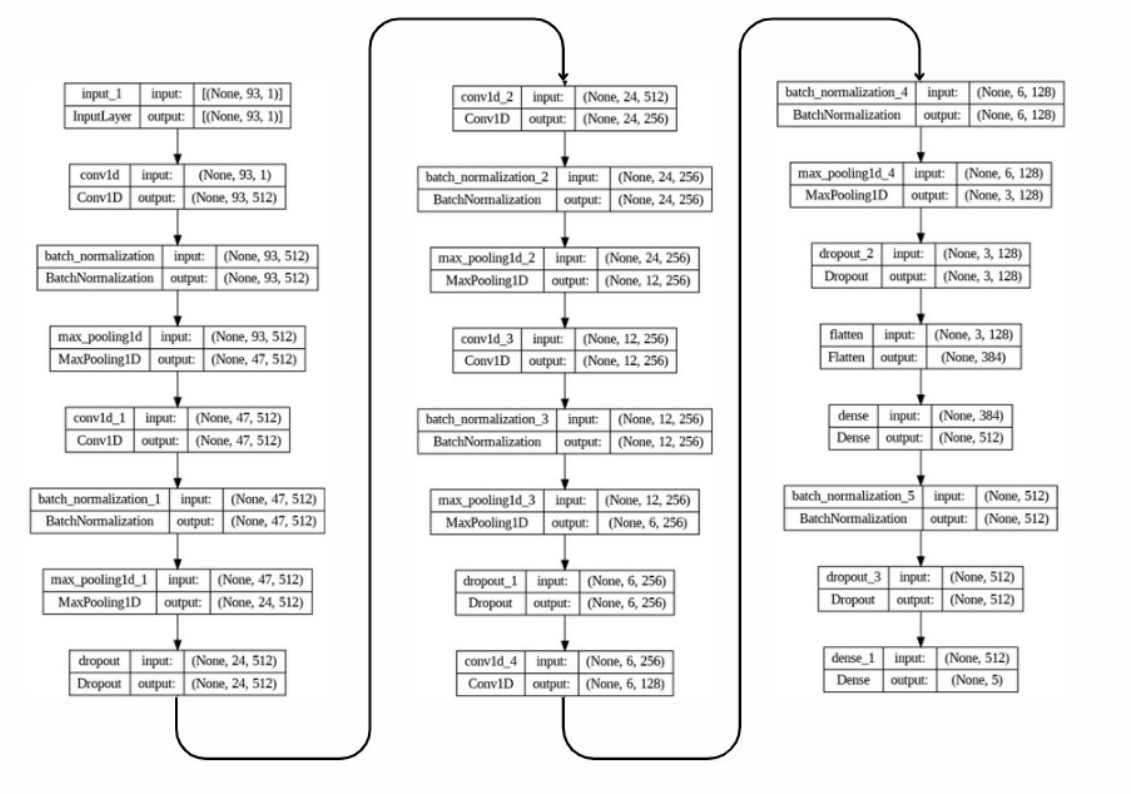

Figure 6.6 Model Architecture

During the training process early stopping, model checkpoint, and reduce learning callback functions were used. The model was trained with training dataset and its performance was evaluated with validation dataset.

Success of the model is evaluated with the accuracy on test data, classification report, and confusion matrix. Classification report consists of precision, recall, F1 score, and support for each emotion class.

7
Performance Analysis

This chapter covers accuracy and performance analysis of the research.

7.1 Accuracy

Performance of the model was evaluated with accuracy, classification report, and confusion matrix. For this step, models ability to recognize different emotion labels was calculated.

Models accuracy and loss values on training and validation sets were checked up throughout the training process.

Change in accuracy and loss values in training and validation sets over time is given in Figure 7.1.

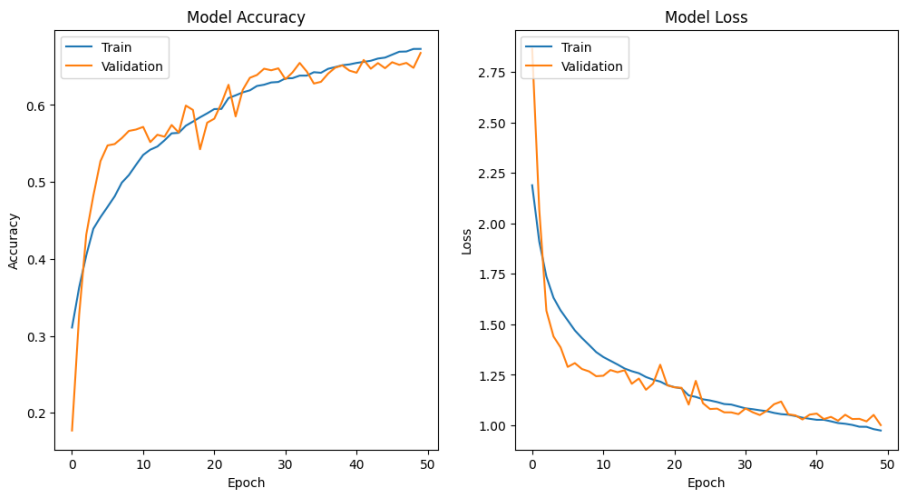

Figure 7.1 Accuracy/Loss of openSMILE

During the training period, accuracy values in the training and validation sets continuously increase and loss values decrease. This indicates that the model performs better with time. Callback functions were essential to reduce the risk of overfitting.

7.2 Performance

Performance of the model was evaluated with precision, recall and F1 score metrics. The result is given in Table 7.1.

Table 7.1 Classification Report

	Precision	Recall	F1-Score	Support
Anger	0.76	0.57	0.65	1483
Happy	0.64	0.54	0.59	1571
Neutral	0.62	0.76	0.68	2454
Sad	0.69	0.72	0.70	1969
Surprise	0.65	0.65	0.65	1330
Accuracy	0.66	0.66	0.66	8807
Macro Average	0.67	0.65	0.65	8807
Weighted Average	0.67	0.66	0.66	8807

Classification table points out the success of the model for every class. The model achieved higher success on 'Neutral' and 'Sad' classes.

Confusion matrix was used for a more enhanced evaluation of the models performance. This matrix is used to determine the classes, where the model makes the most mistakes. The confusion matrix is given in Figure 7.2.

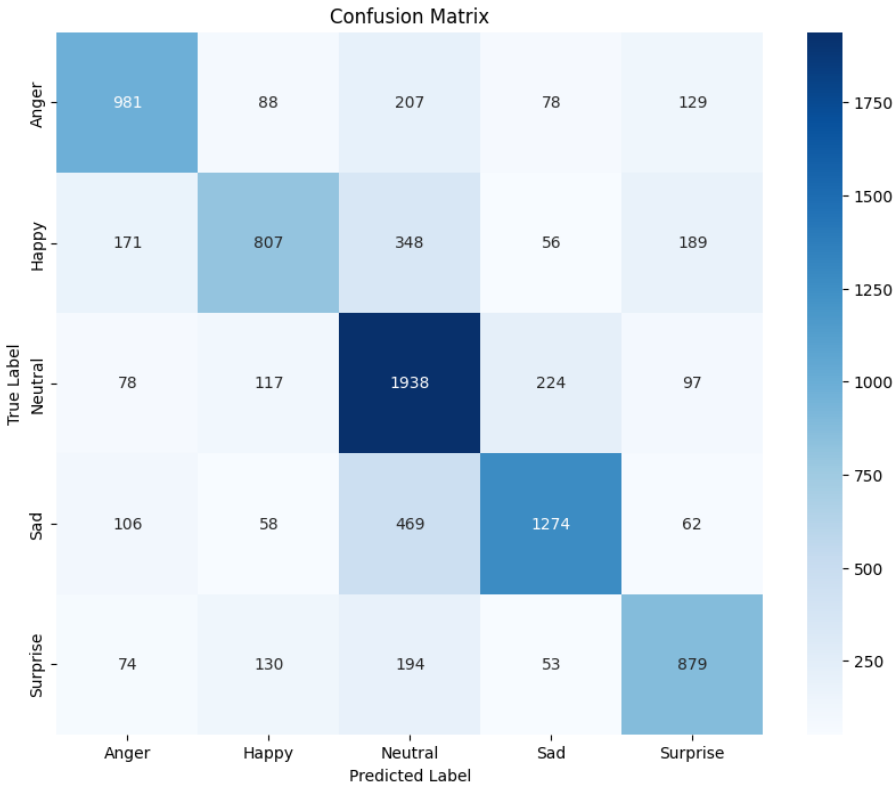

Figure 7.2 Confusion Matrix of openSMILE

7.3 Test Dataset Success

After evaluating the model on test dataset we have achieved 0.69 success rate, compatible with the training and validation rate. This score shows that the model performs successfully in the terms of multimodal emotion recognition.

8
Experimental Results

Example outputs from the system are given below. The output shows the file path of the audio file, text spoken in the audio, predicted emotion, and true emotion.

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/test_dataset/savee/DC_a04.wav	who authorized unlimited expense account	Anger	Anger

Figure 8.1 Example of Anger 1

Filepath	Text	Predicted Emotion	True Emotion
233,/content/drive/MyDrive/bitirme/test_dataset/savee/JE_su03.wav	will you tell me why	Surprise	Surprise

Figure 8.2 Example of Surprise 1

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/test_dataset/savee/JK_h13.wav	he would not carry a briefcase	Happy	Happy

Figure 8.3 Example of Happy 1

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/sound_datasets/crema_dataset/AudioWAV/1058_TAI_NEU_XX.wav,Crema_1058	The airplane is almost full	Neutral	Neutral

Figure 8.4 Example of Neutral 1

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/test_dataset/savee/DC_sa12.wav	American newspaper reviewers like to call his place nihilistic	Sad	Sad

Figure 8.5 Example of Sad 1

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/MELD-RAW/MELD.Raw/train/train_splits_mp3/dia1016_utt16.mp3,Meld_Ross	"No, I'm not gonna pick them up."	Anger	Anger

Figure 8.6 Example of Anger 2

Filepath	Text	Predicted Emotion	True Emotion
40230,/content/drive/MyDrive/bitirme/IEMOCAP_full_release/Session1/sentences/wav/merged/Ses01M_impro03_M004.wav,Ses01M_impro03_M004	"No, no, no, no, no, no she just... she's not a citizen but she wants her visa, yeah. She's going to pay me fifty thousand dollars to marry her and stay married to her for two years."	Surprise	Surprise

Figure 8.7 Example of Surprise 2

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/test_dataset/savee/JK_h13.wav	he would not carry a briefcase	Happy	Happy

Figure 8.8 Example of Happy 2

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/MELD-RAW/MELD.Raw/train/train_splits_mp3/dia959_utt0.mp3,Meld_Ross	"I didn't even realise how late it was, until I noticed the 5 o'clock shadow on her head."	Neutral	Neutral

Figure 8.9 Example of Neutral 1

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/test_dataset/savee/KL_sa06.wav	before Thursday exam review every formula	Sad	Sad

Figure 8.10 Example of Sad 2

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/sound_datasets/esd/Emotion Speech Dataset/0011/Angry/0011_000382.wav,Esd_0011	Who is been repeating all that hard stuff to you?	Anger	Anger

Figure 8.11 Example of Anger 3

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/MELD-RAW/MELD.Raw/train/train_splits_mp3/dia42_utt1.mp3,Meld_Guru Saj	"Hello, I am Guru Saj--Whoa!! That's supposed to be a duck right? 'Cause otherwise, this is waaay out of my league."	Surprise	Surprise

Figure 8.12 Example of Surprise 3

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/MELD-RAW/MELD.Raw/train/train_splits_mp3/dia108_utt2.mp3,Meld_Rachel	Hey! Great advice on that Joey thing!	Happy	Happy

Figure 8.13 Example of Happy 3

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/MELD-RAW/MELD.Raw/train/train_splits_mp3/dia62_utt11.mp3,Meld_Chandler	But this is eleven. It's almost twice as hard up here..,	Neutral	Neutral

Figure 8.14 Example of Neutral 3

Filepath	Text	Predicted Emotion	True Emotion
/content/drive/MyDrive/bitirme/IEMOCAP_full_release/Session3/sentences/wav/merged/Ses03F_impro02_F008.wav,Ses03F_impro02_F008	And Uh- Yeah. Uh- They're going to need um- more nurses on call for uh- to go overseas to help out with the war.	Sad	Sad

Figure 8.15 Example of Sad 3

9
Result

This study consists of data preprocessing, feature extraction from text and audio, creation of an integrated model and training the model. The system, created as a result of this study, takes the audio and text data as an input and predicts the label of the given data as an output based on the training information.

Dataset used for this reasearch consists of audio and text data taken from various datasets. These datasets create a broad and diverse emotional base and provide the necessary data for training the model.

The system extracts features from audio data and combines them with text data to determine the dominant emotion. Throughout this process openSMILE library, TF-IDF vectorization, and SVC are used. Lastly, a CNN model is used to create the output. The model trained and used in this project achieved a high level of success with accuracy of 0.67. Results obtained from train and test datasets indicates that the model can successfully recognise different emotion classes. These emotions consist of anger, happiness, neutral, sadness, and surprise.

The model developed in this study can be used in various areas such as, customer service, social media management, emergency call centers and much more.

To sum up, this research focuses on developing a system that analyze and recognize emotions using audio and text data. According to the evaluation, the model can successfully recognize and predict emotions. The system can be used in different application areas and can be developed further.

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Project System Informations

System and Software: Microsoft Windows 11, Apple macOS 14.5, Python
Required RAM: 15GB
Required Disk: 256MB